



Pharmacological and immunomodulatory modes of action of medically important phytochemicals against arthritis: A molecular insight

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Abstract

Arthritis is a common illness that affects joints and it may result in inflammation and pain. Even though arthritis usually affects older people, it can also affect children, adults, and both genders. Numerous arthritic mouse models have been developed but the CIA model of rheumatoid arthritis (RA) has received the most attention. With the use of steroids, DMARDs, and NSAIDs, therapy objectives such as reduced disease incidence and better pain management are achieved. Long-term usage of these therapeutic approaches may have negative side effects. Herbal medications are the source of several medicinal substances. Studies have explored the potential benefits of medicinal plants in treating RA. These benefits include up-regulating antioxidant potential, inhibiting cartilage degradation, down-regulating inflammatory cytokines such as NF- κ B, IL-6, and TNF- α , and suppressing oxidative stress. In this review, we systematically discuss the role of traditional medicinal plants in rheumatoid arthritis (RA) disease treatment. The role of different medicinal plants such as *Curcuma longa*, *Syzygium aromaticum*, *Zingiber officinale* and *Withania somnifera*, against arthritis is discussed in this review.

Keywords Rheumatoid arthritis · Medicinal plants · Mechanism · Phytochemicals

Abbreviations

AKBA	3-O-acetyl-11-keto-boswellic acid	FCA	Freund's Complete Adjuvant
CBP	Peptide that binds to collagen	HNE	4-hydroxy 2-nonenal
CFA	Chartered Financial Analyst	IFN- γ	Interferon- γ
CRP	C-reactive protein	IgA	Immunoglobulin A
CIA	Collagen-induced arthritis	IgG	Immunoglobulin G
CII	Collagen type II	IL-1	Interleukin-1
COX	Enzyme cyclooxygenase	IL-6	Interleukin-6
DCs	Dendritic cells	KBA	11-keto-boswellic acid
DMARDs	Disease-modifying anti-rheumatic drugs	LPS	Lipopolysaccharide
ERK1/2	Extracellular signal-regulated protein kinases 1 and 2	MAPK	Mitogen-activated protein kinases
		MHC	Major histocompatibility complex
		MMPs	Matrix metalloproteinase
		MTX	Methotrexate
		MW	Molecular weight
		NF- κ B	Nuclear factor-kappa B
		NK cells	Natural killer cells
		NSAIDs	Non-steroidal anti-inflammatory agents
		PGE2	Prostaglandin E2
		RA	Rheumatoid arthritis
		SCW	Streptococcal cell wall
		SFs	Synovial fibroblasts

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TCM	Traditional Chinese medicine
TGF	Transforming growth factor
THP-1	Tohoku Hospital Pediatrics-1(a cell culture of human-derived leukemia monocytic cells)
TNF- α	Tumor necrosis factor-alpha
TQ	Thymoquinone

Introduction

Musculoskeletal problems are the second most typical reason for body impairment, globally [1] and pose a danger to healthy aging [2]. The worldwide disability load linked to the musculoskeletal system is strongly impacted by arthritis [3]. More than 100 distinct arthritic conditions, which are agonizing, chronic, and a collection of disorders, are collectively referred to as “arthritis” [4]. An acute or chronic disease that affects one or more joints is also known as arthritis. Its clinical spectrum may include stiffness, discomfort, and joint abnormalities in addition to joint swelling [5]. It can also manifest in forms, such as osteoarthritis, psoriatic arthritis, and RA [6]. One of the most common chronic inflammatory illnesses is rheumatoid arthritis [7]. Symmetric polyarthritis, a symptom of rheumatoid arthritis that affects the synovial membrane and has been linked to high levels of IgM and IgA antibodies, is the disease’s distinguishing feature [8]. Approximately 1% of people have rheumatoid arthritis (RA) all over the world, with men being less likely than women to develop it [9].

Rheumatoid arthritis mechanism of action

Genetic, environmental, and autoimmune variables are the key causes of RA’s pathogenesis [10]. The main causes of RA in people with genetic susceptibility include smoking, hormones, infections, microbiome, and other variables [11]. Antigen-activated CD4⁺ T cells are initiated by the different environmental and genetic variables causing the disease through the immune cells activation, such as synovial fibroblasts, macrophages, and monocytes [12]. TNF- α , Interleukin-6, and interleukin-1 are produced as a result of CD4⁺ T-cell activation, which is the primary mediator of RA [13]. These pro-inflammatory mediators have a crucial role in the RA pathogenesis, including articular manifestation through inflammatory infiltration increase of cell types, mainly macrophages, B lymphocytes, T cells and bone erosion leading to the formation of autoantibodies and the release of Interleukin-6 and TNF- α as shown in Fig. 1 [14].

Autoimmune responses in rheumatoid arthritis cause macrophages and T cells to release inflammatory cytokines variety in the joints. This causes pain and swelling, ongoing

bone and cartilage degradation, and systemic consequences over time that raise cardiovascular morbidity and mortality rates [15]. Additionally, these cytokines prevent collagen breakdown and proteoglycan production. Proteolytic enzyme activation like collagenases and MMPs causes cartilage degradation [16]. Additionally, certain protein molecules, such as cytokines and chemokines, cause synovitis and tissue damage by luring lymphocytes, monocytes, and neutrophils as well as the COX-2 enzyme. This tissue damage results in synovial joint hyperplasia and pannus formation that causes apoptosis in synovial fibroblasts [17]. The mechanism of RA within the joint capsule is also described below:

Since cartilage lacks blood vessels and nerves and is primarily made up of cartilage cells, extracellular matrix and fibers, it receives nutrition from blood vessels through the perichondrium via nutrient penetration into the intercellular substance [18]. Chondrocytes, the primary cartilage cells, control the homeostasis and cartilage’s dynamic stability during catabolism and anabolism [19]. Any disruption of this equilibrium may cause damage to the cartilage tissue, which could result in arthritis. These internal and external causes may include inflammatory environments [20]. Immune cells such as natural killer (NK) cells, dendritic cells (DCs), macrophages, neutrophils, T lymphocytes, and B lymphocytes are the main producers of IL-1 as shown in Fig. 2 [21].

It is reported that the expression of the collagen type II (COL2A1) and chondrocyte-specific genes *Sox9* can be downregulated by IL-1, a process that may be sped up by T cells secreting IL-3 [22]. On the other hand, T cells’ production of IL-6 and IL-3 protects the cartilage by preventing the development of MMPs through the use of tissue inhibitors. In some circumstances, IL-6 prevents the formation of proteoglycans in chondrocytes and increases IL-1 to cause proteoglycan denaturation [23]. IL-6, thus, plays an important role in protecting and destroying cells of cartilage, both of which are solely controlled by the immune system’s reaction to allergens [24]. While transforming growth factor (TGF) is synthesized by T cells in comparison to other cells, and it promotes cartilage remodeling by down-regulating the expression of Interleukin-6 receptor in chondrocytes, thereby preserving the homeostasis of the cartilage and up-regulating the differentiation-related transcription gene *sox9* [25].

Although the specific origin of RA is still unknown, it is thought that the pathophysiology of RA is regulated by B cells, T cells, excess production of several proinflammatory cytokines, and dysregulation of anti-inflammatory cytokines [26]. Different animal models have been used to evaluate potential anti-arthritis drugs for clinical use. For

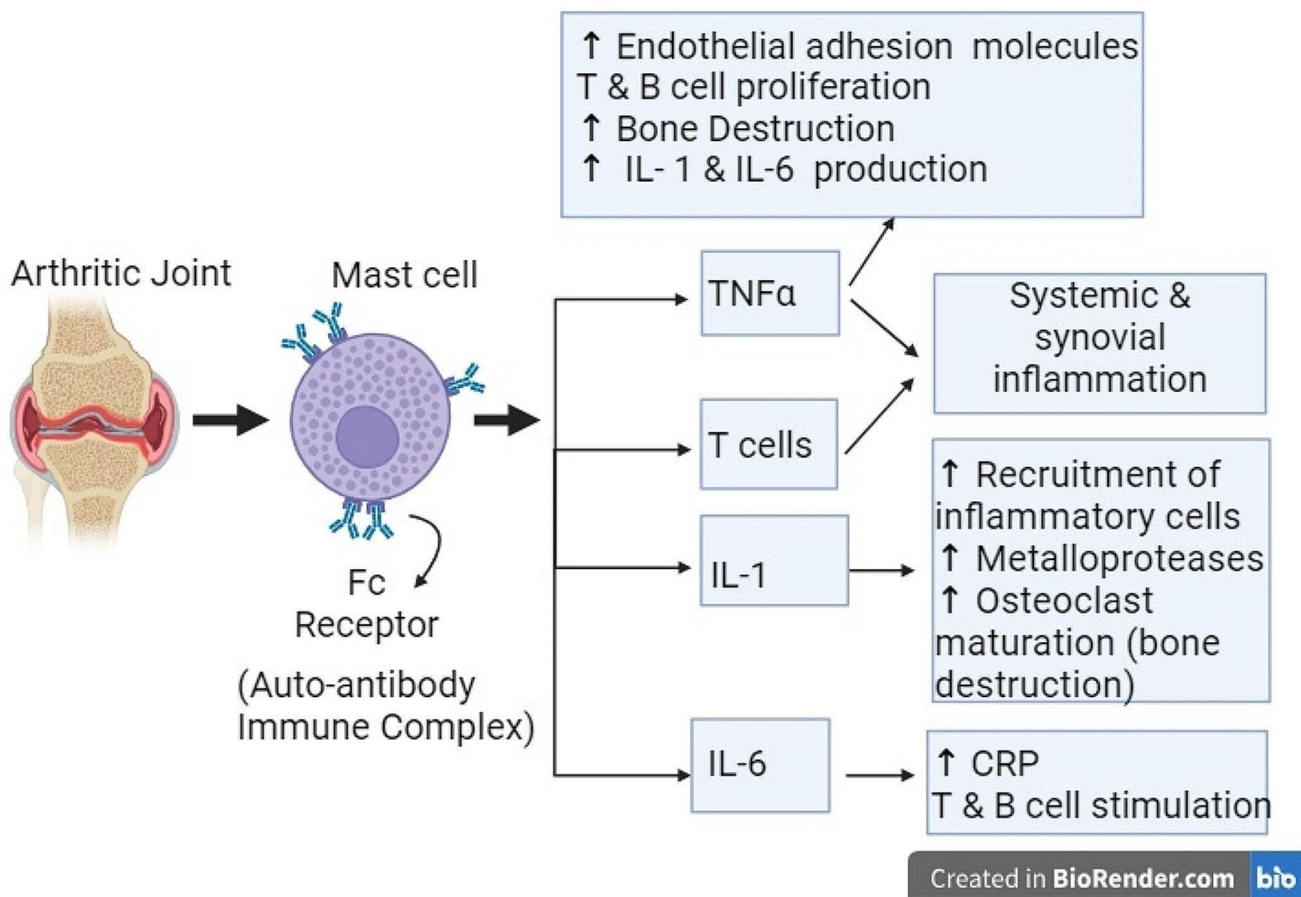


Fig. 1 Rheumatoid arthritis is caused by autoimmune reactions that cause macrophages and T cells to release a variety of inflammatory cytokines into the joints. This causes swelling and pain, gradually deteriorating bone and cartilage, and systemic consequences that even-

this purpose, collagen is widely used to induce arthritis in animal models.

Collagen-induced arthritis

Cartilage autoimmune response and the development of erosive polyarthritis were observed in rats that were immunized for the first time in 1977 with a chicken, rat, or human type II collagen (CII) emulsion in complete Freund's adjuvant (CFA) [27]. In 1986 and 1980, respectively, mice and monkeys were used to recreate this CIA model. A popular animal model for examining the effects of novel treatments for rheumatoid arthritis is collagen-induced arthritis. An injection of a collagen emulsion induces CIA [28], and with the antibody formation and activation of B-cell and T-cell activation, an immune response is initiated that targets collagen [29]. The local autoimmune response is triggered when antibodies identify the collagen in the joints. Consequently, cytokines and other inflammatory mediators are produced [30].

tually raise the risk of cardiovascular disease and death. The pathophysiology of RA is more complicated than is depicted in terms of the interactions between cytokines and other cell types

Future expectations for RA prevention are shaped by the discovery of a preclinical stage, expanding knowledge of the natural history and mechanisms of RA development, and novel therapeutic approaches [31]. Different therapeutic approaches are described below.

Treatments of RA and complications associated with pharmacotherapy

Commonly, NSAIDs and steroidal anti-inflammatory drugs (corticosteroids) are used to reduce pain and inflammation in RA patients. Medication for RA includes B-cell depleting rituximab, IL-1 antagonist tocilizumab, TNF-blockers (etanercept, infliximab, adalimumab, and certolizumab), IL-6 antagonist tocilizumab, T-cell co-stimulator abatacept, and IL-1 receptor antagonist [32]. In addition, biological therapies and disease-modifying antirheumatic drugs (DMARDs) are used to reduce the progression of damage and to prevent the onset of illness [33]. Moreover, RA can be treated with other drugs including cyclosporine,

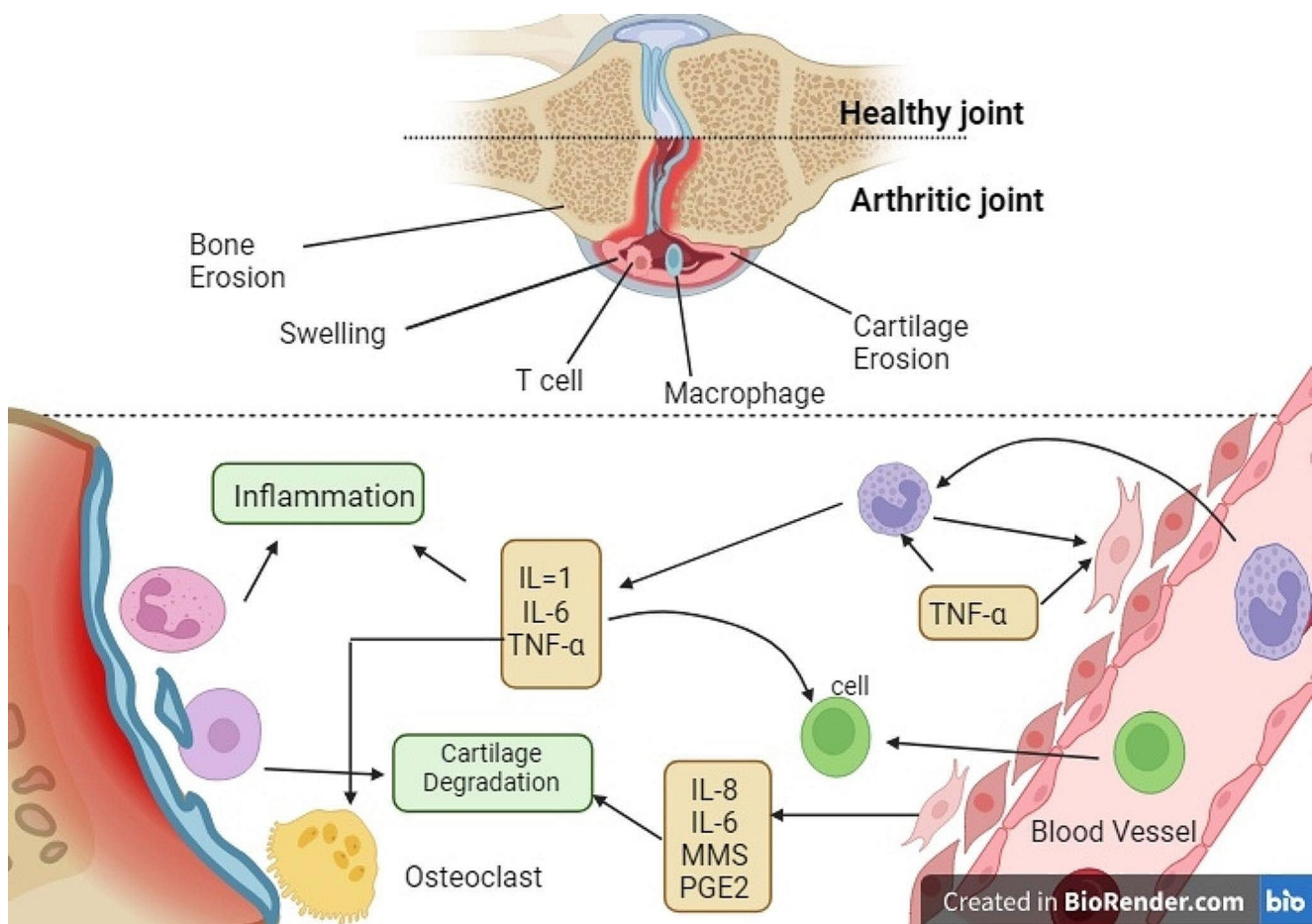


Fig. 2 Mechanism of RA within the joint capsule. TNF- α affects synoviocytes, monocytes/macrophages and osteoclasts; initiating RA pathogenesis. Synoviocytes line the capsule and T-cells infiltrate the synovial membrane, initiating inflammation via TNF- α . Joint degradation

occurs via macrophage recruitment and secretion of inflammatory cytokines. Bone erosion occurs by osteoclasts and inhibition of collagen secretion by synoviocytes

minocycline, leflunomide, azathioprine, myochrysine, sulfasalazine, gold, hydroxychloroquine, and methotrexate [33, 34]. However, drugs used to treat arthritis may impact fertility and pregnancy. Using anti-rheumatoid medications like methotrexate, azathioprine, and cyclophosphamide to treat RA during pregnancy has proven to be a challenging approach. These medications can cause a number of potentially fatal teratogenic side effects, such as abnormalities in the developing fetus, problems with growth, and an increased risk of preterm birth. The focus of modern therapy has thus switched to herbal remedies that are complementary and alternative and have little adverse effects [35]. Several medicinal plants are used to make herbal medications. The active ingredients found in herbal plants, such as flavonoids, carotenoids, phenols, terpenes, and sesquiterpene lactones, are advantageous because they have anti-inflammatory and antioxidant properties. These characteristics make them a potent therapeutic strategy for the treatment of RA [34, 36].

Medicinal plants for rheumatoid arthritis treatment

Plant-based medications with the ability to control the expression of pro-inflammatory signals have the potential to treat arthritis [37]. Investigations into the anti-inflammatory properties of medicinal plant extracts have revealed a variety of mechanisms, some of which include blocking pro-inflammatory mediators [38].

Clinical studies have demonstrated the effectiveness of plant-derived alkaloids such as colchicines, isoquinoline, and aconitine as an arthritis treatment by lowering pain symptoms and inflammation [13]). The colchicines block neutrophil emigration, reduce activation of inflammasome and further reduce the levels of leukotriene and cytokines, which appear to be higher in arthritis characterized by chronic inflammation [39]. By inhibiting the enzyme cyclooxygenase (COX), the other alkaloids (indole, aconitine, and isoquinoline), terpenoids, and flavonoids demonstrate

antipyretic and analgesic effects by reducing inflammatory mediators and preventing joint degeneration [35].

Important mechanisms of natural medicines in the treatment of RA include reduction of the expression of inflammatory mediators, immunological regulation, inhibition of osteoclast development, anti-oxidative stress, and inhibition of proliferation and migration of FLSs. Fig. 3 summarizes the primary signaling pathways and transformation factors: PI3K/Akt, NF- κ B, JAK2/STAT3 signaling, MAPK, AP-1, Nrf2, HIF-1 α , and SIRT1 that are targeted by bioactive compounds of different medicinal plants used to treat RA.

When used in conjunction with MTX for the treatment of RA, several natural medications, including steroidal lactones, 8 shogaol, desmethoxycumin, curcumin, thymoquinone, eugenol and eugenol acetate, can produce greater therapeutic benefits and/or fewer side effects. These demonstrate in full the benefits of using natural medications to treat RA: they are safe, effective, natural, and have little harmful or adverse consequences [11, 40].

Curcuma longa

Originally from India, the perennial shrub *Curcuma longa*, a member of the Zingiberaceae family, is now grown in Sri Lanka, China, and other tropical countries [41]. Studies have revealed that curcuminoids including curcumin, desmethoxycurcumin, bisdemethoxycurcumin, monodemethoxycurcumin, dihydrocurcumin, and cyclocurcumin are present in turmeric, one of the plants that has been the subject of the most research [42]. The major ingredient in turmeric, curcumin, has drawn a lot of interest as a plant-based substance with pharmacological benefits that are pleiotropic. It has immunomodulatory, neuroprotective, hypoglycemic, antioxidant, anti-inflammatory, and antibacterial properties [43]. The pre-clinical experiment investigated how giving animals a low dose of *C. longa* extract prevented both the chronic and acute phases of arthritis and found that curcuminoid, the active element in *C. longa*, has potent anti-inflammatory activities [13].

By preventing the generation of pro-inflammatory cytokines including Interleukin-1, TNF- α and Interleukin-6, curcumin is successful in preventing the advancement of

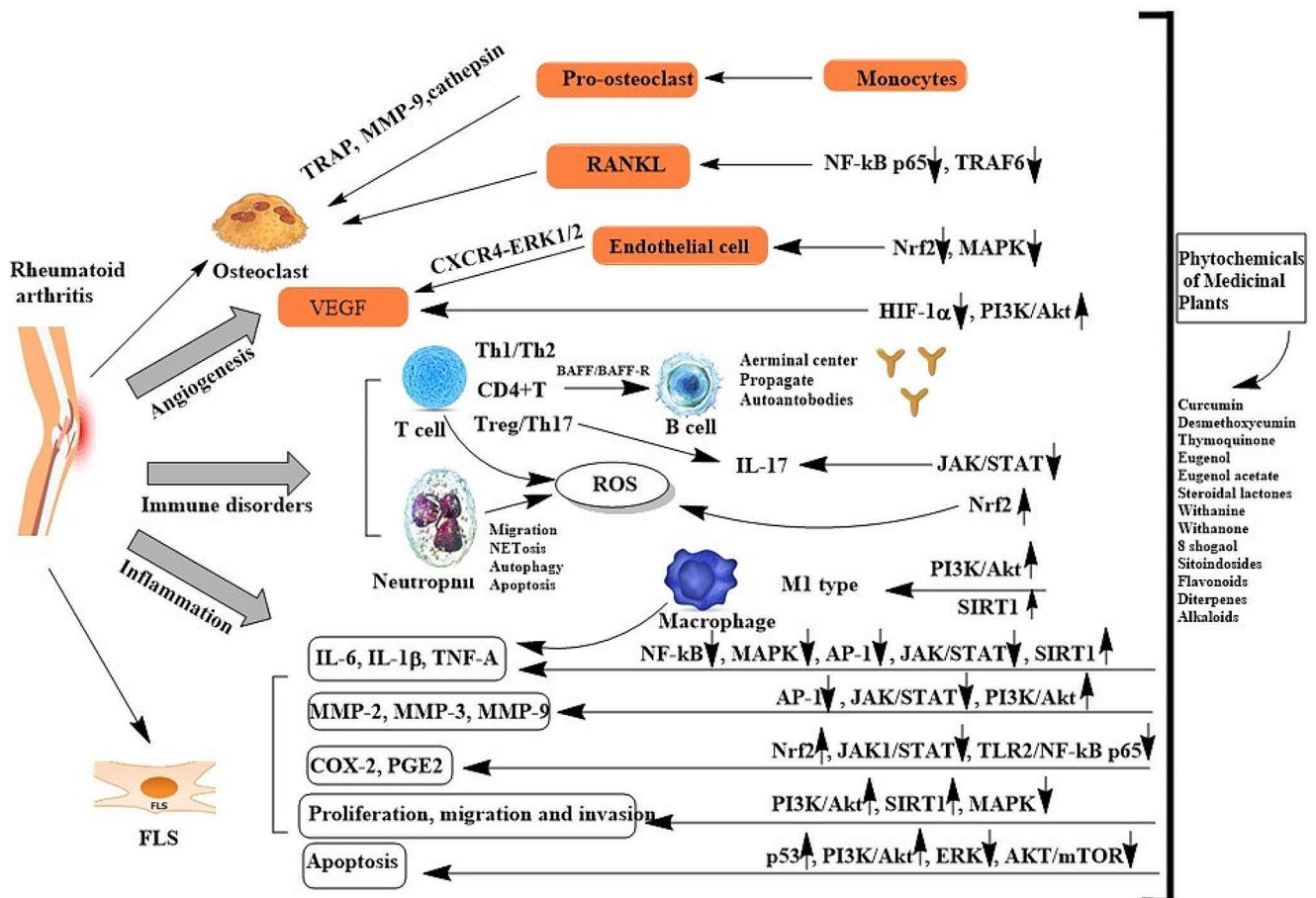


Fig. 3 Mechanisms of natural medicines (phytochemicals) against RA. ↑: increase, promote or up-regulation; ↓: decrease, down-regulation or inhibit

arthritis in the rat model [44], as shown in Fig. 4. Adipokines like leptin, adiponectin, and adiponectin are suppressed by curcumin nanoparticles when applied to the mouse knee. To prevent cartilage deterioration, curcumin nanoparticles have also shown anti-inflammatory capabilities [45]. Curcumin has been discovered to be responsible for enhancing the anti-arthritic properties of other therapeutic plants, including black pepper and sarsaparilla root [46].

Nigella sativa

Nigella sativa, originally native to Southwest Asia, Southern Europe, and North Africa, is now grown in various countries, including the Middle Eastern Mediterranean region, Saudi Arabia, Pakistan, Southern Europe, Syria, India, and Turkey [47]. Due to the presence of its main phytochemical, thymoquinone, *N. sativa*'s oil and seeds have been claimed to have anti-inflammatory properties [48]. The average composition of *N. sativa* oil is 0.5–1.6% essential fatty acids, 22.7% protein, 35.6–41.6% fat (oil), and 32.2% carbohydrate. Linoleic acid makes up the majority of fixed oil in this plant (approximately 57.3%) [49].

Using an animal model of experimentally induced arthritis, the anti-RA effects of *N. sativa* and its active component TQ were examined [50]. According to a study, taking TQ orally for 21 days at a dose of 5 mg/kg per day reduced levels of inflammatory cytokines (tumor necrosis factor- α , interleukin-6, interferon- γ , interleukin-1, prostaglandin E2 [PGE2]) and oxidative stress markers (myeloperoxidase, articular elastase, nitric oxide, and lipoxygenase) [49]. In a study, it was discovered that thymoquinone reduced the levels of NF- κ B, IL-1, and Tumor necrosis factor- α in arthritic rats given Freund's Complete Adjuvant (FCA) [51]. Therefore, *N. sativa* can be used as a useful candidate in arthritis management. Additionally, research using human RA fibroblast-like synoviocytes in cell culture demonstrated that thymoquinone prevented the proliferation of synoviocytes induced by lipopolysaccharide (LPS), the production of 4-hydroxynonenal induced by H₂O₂ [52], and the levels of IL-1, TNF- α , metalloproteinase-13, cyclooxygenase-2, and prostaglandin E2. Thymoquinone inhibited p38, MAPK, ERK1/2, and NF- κ B p65 phosphorylation caused by LPS in a time-dependent manner. Additionally, research has demonstrated that giving arthritic rats thymoquinone orally lowers the serum levels of HNE, TNF- α , and Interleukin-1 [53].

Syzygium aromaticum

Syzygium aromaticum, popularly known as clove, is a dried flower bud that is endemic to the islands of Indonesia, but has recently spread around the world [54]. Several studies

have demonstrated the immunomodulatory and regenerative properties of clove oil and its constituents for the treatment of rheumatoid arthritis (RA) [55, 56]. Due to the inclusion of eugenol acetate, β -caryophyllene, and eugenol (EU) as the main components, studies have shown that clove has excellent antibacterial, analgesic, antioxidant, and anesthetic qualities [56].

Eugenol, which makes up 45–90% of clove oil, is responsible for the anti-inflammatory benefits of cloves by blocking the nuclear factor- κ B signaling pathway and preventing the production of several cytokines [57]. Additionally, by preventing the generation of cytokines in macrophages, eugenol dimers demonstrated chemopreventive effects. Eugenol may also have anti-inflammatory effects on arthritis, making it a potential treatment against the disease [58].

Withania somnifera

W. somnifera is a xerophytic plant that nurtures abundantly in Africa, Pakistan, the Mediterranean, Sri Lanka, and India [59]. Ashwagandha, also known as *Withania somnifera*, is a plant in the Solanaceae family that is extensively used in Ayurvedic medicine to treat different conditions, such as cardiovascular disease, anxiety, arthritis, asthma, amnesia, hepatitis, and neurological illnesses [60]. *W. somnifera* found to contain steroidal lactones known as withanolides that are pharmacologically active. Alkaloids from the plant's roots known as withanine make up 38% of the total weight of all known alkaloids [60]. In the form of leaves, roots, flowers, roots, and stems, 29 common metabolites produced from the leaf and root extracts have medicinal value [61]. With a long history that dates back to the year 6,000, *W. somnifera* has been utilized medicinally. The plant's primary metabolites include withanolides, including withanolide A, withanone, and withaferin A, as well as withanosides, sitoindosides, and other alkaloids that may have therapeutic and medicinal uses [62]. *W. somnifera* has anti-arthritic qualities because it inhibits inflammatory chemicals and increases antioxidant activity [63]. Additionally, it has been noted that *W. somnifera* extract reduces the collagenolytic activity, which further inhibits collagen degradation [46]. Several clinical studies have also been conducted to determine the effectiveness of *W. somnifera* extracts in treating joint pain, and the extracts are successful in treating arthritic conditions [64].

Zingiber officinale

Ginger is a key ingredient in different alternative medicine systems and is one of the known culinary and medicinal ingredients in the world [65]. It is effective against arthritis, rheumatological diseases, muscular aches, headaches,

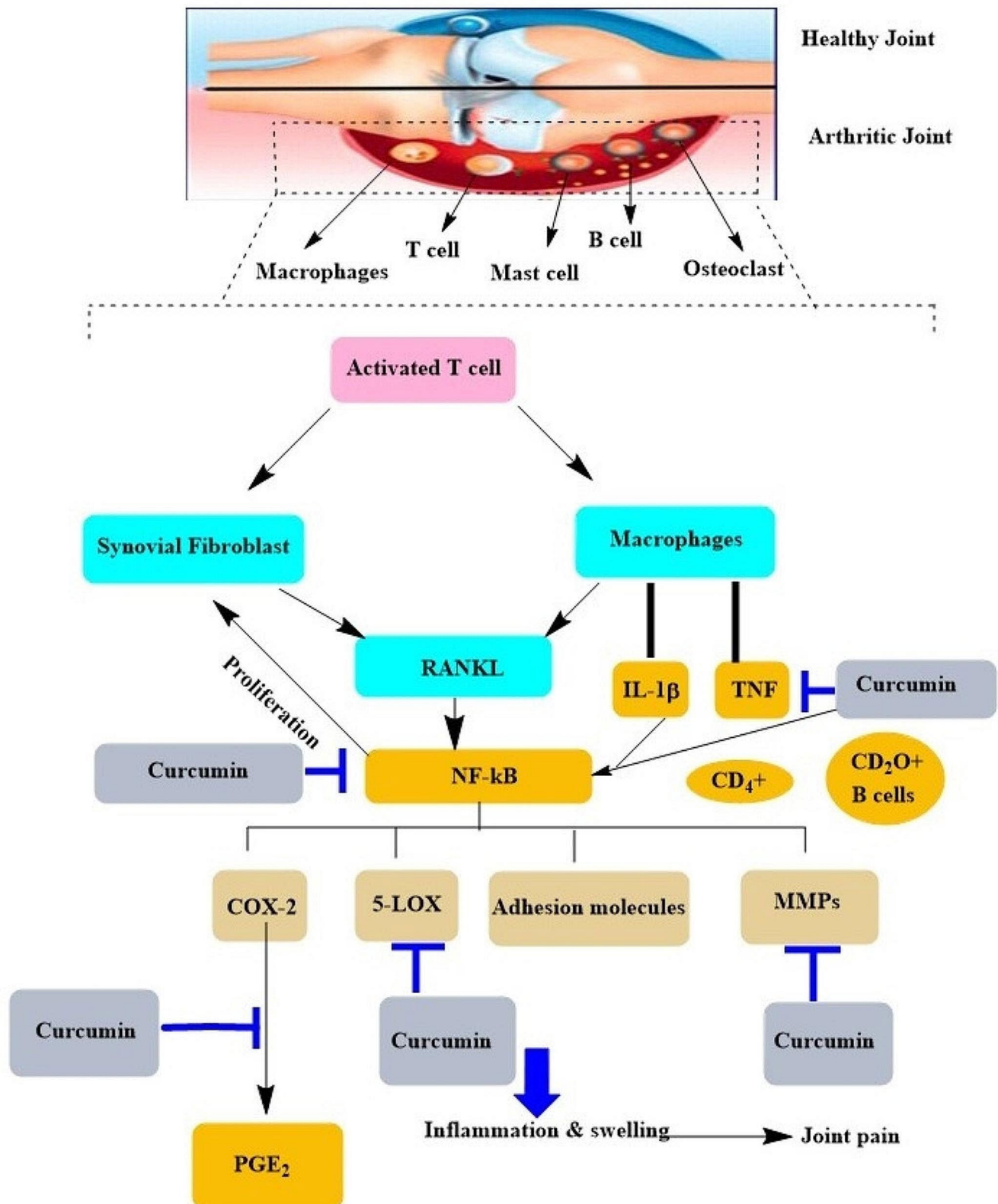


Fig. 4 Immunomodulatory effects of curcumin on dysregulated immune cells in RA

nausea, stomach distress, diarrhea, and colds. It also acts as a carminative and an antifatulent [66]. In both preclinical and clinical investigations, ginger has received extensive attention for its anti-arthritis properties. Rats' paws and joint swelling brought on by *Mycobacterium tuberculosis* are significantly reduced by ginger oil. The ginger extract in alcohol was efficient in lowering inflammation and collagen-induced arthritis in rats [43].

By upregulating the expression of the forkhead-box-P3 (FoxP3) gene and downregulating the expression of T-bet and retinoic acid receptor-related orphan nuclear receptor gamma (ROR γ t) genes, ginger (rhizomes of *Zingiber officinale*) improved RA as shown in Fig. 5 [67]. FoxP3 expression promotes the growth and functionality of regulatory T (T-reg) cells, as it is a crucial transcription factor for T-reg cells [68]. TGF- β and IL-10, two immunomodulatory cytokines, are produced when T-reg cells are activated, and they also lessen inflammation [69]. It is widely believed that ROR γ t is a key regulator of Th17 cell differentiation and IL-17 cytokine production, both of which are critical for the onset of several autoimmune diseases, including arthritis.

Immune cell transcription factor T-bet is said to control the generation of the pro-inflammatory cytokine IL-1 α in dendritic cells [70]. Both of these transcription factor genes combined expression levels are declined, which may aid in lessening RA [71].

Mechanistic studies demonstrate that the therapeutic actions of ginger or its phytochemicals are mediated by scavenging free radicals [72], increasing antioxidant enzymes and antioxidant molecules, reducing the leukocyte infiltration (including macrophages /monocytes and lymphocytes) into the synovial cavity of the knee [73], inhibiting iNOS and nitric oxide anti-inflammatory activity, and by suppressing prostaglandin synthesis through inhibition of cyclooxygenase-2 and -1 [43].

8-Shogaol had strong inhibitory effects on TNF- α , IL-6, and IL-8-mediated inflammation and migration [74]. 8-Shogaol directly and specifically reduced TAK1 activity, which in turn repressed the MAPK, Akt, and IKK signaling pathways as shown in Fig. 6. Furthermore, in the adjuvant-induced arthritic (AIA) rat model, 8-shogaol therapy decreased paw thickness and enhanced walking ability [75].

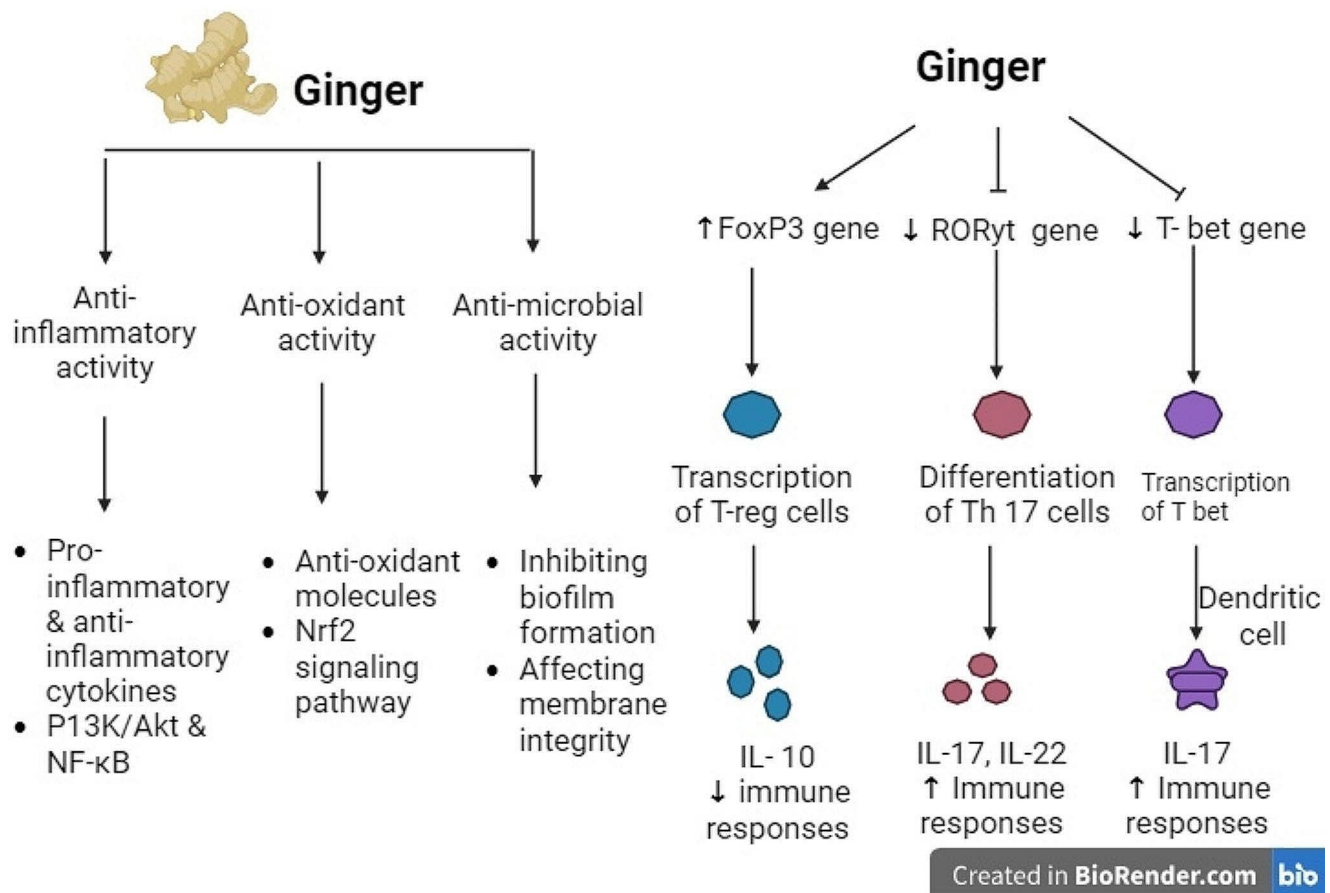


Fig. 5 Biological activities and simplified mechanisms of the immunomodulatory effects of *Zingiber officinale*. Abbreviations—T-reg cells: regulatory T-lymphocytes; Th17: helper T-lymphocyte 17; IL:

interleukin; FoxP3: forkhead-box-P3; ROR γ t: retinoic-acid-receptor-related orphan nuclear receptor gamma

By boosting the anti-inflammatory cytokines production while lowering the production of pro-inflammatory cytokines and activating the antioxidant system, ginger has been discovered to have antiarthritic properties. *Z. officinale* also has strong anti-inflammatory and anti-rheumatic capabilities [76].

Tinospora cordifolia

Tinospora cordifolia, also known as Amruta in Ayurveda, has great therapeutic value as well as effective analgesic, anti-inflammatory, and immunomodulatory qualities [46]. The anti-inflammatory and antioxidant activity of *T. cordifolia* was examined in activated human monocytic (THP-1) cells because the traditional use of the plant has been scientifically supported using animal models for numerous disorders [77]. By interacting with endothelial cells, which later develop into macrophages in the artery lumen, the model system, which is activated by external stress factors at early stages of inflammation, plays a crucial role in generating vascular inflammation and numerous inflammatory disorders [78].

Additionally, *T. cordifolia* is said to inhibit NF- κ B in THP-1 cells, demonstrating its anti-inflammatory capabilities [77]. *T. cordifolia* extract also has been shown to have

anti-arthritic characteristics by reducing the release of chemokines and cytokines such as Interleukin-1, TNF- α , Interleukin-17, and Interleukin-6 in a study employing the rat adjuvant-induced arthritis model of human RA [46, 79].

Commiphora wightii

The Buseraceae plant, *Commiphora wightii*, which grows in northern India and is often referred to as Guggulu in Ayurveda, produces a fragrant resin that has been used to cure a variety of ailments, including cardiac illnesses, inflammation, high cholesterol, arthritis, and obesity [80, 81]. There are other names for it, including guggula, guggul, guggal, gugar, and Indian bdellium [82]. Guggul resin is mostly made up of volatile oil, which is rich in sesquiterpenoids and terpenoids including monoterpenoids, as well as additional terpenoids (di- and triterpenoids), steroids [83], guggultetrols, flavonoids, lignans, sugars, and amino acids [84]. Inhibiting the synthesis of inflammatory molecules like TNF- α , interleukin-1, interleukin-6, and interleukin-23 as well as iNOS and COX-2, has been shown to prevent cartilage deterioration by a lipid-rich extract isolated, guggulipid from *C. wightii* [85]. In an in vivo Freund's full adjuvant-induced arthritic rat model, guggulipid has additionally shown a strong ability to diminish the oxidative

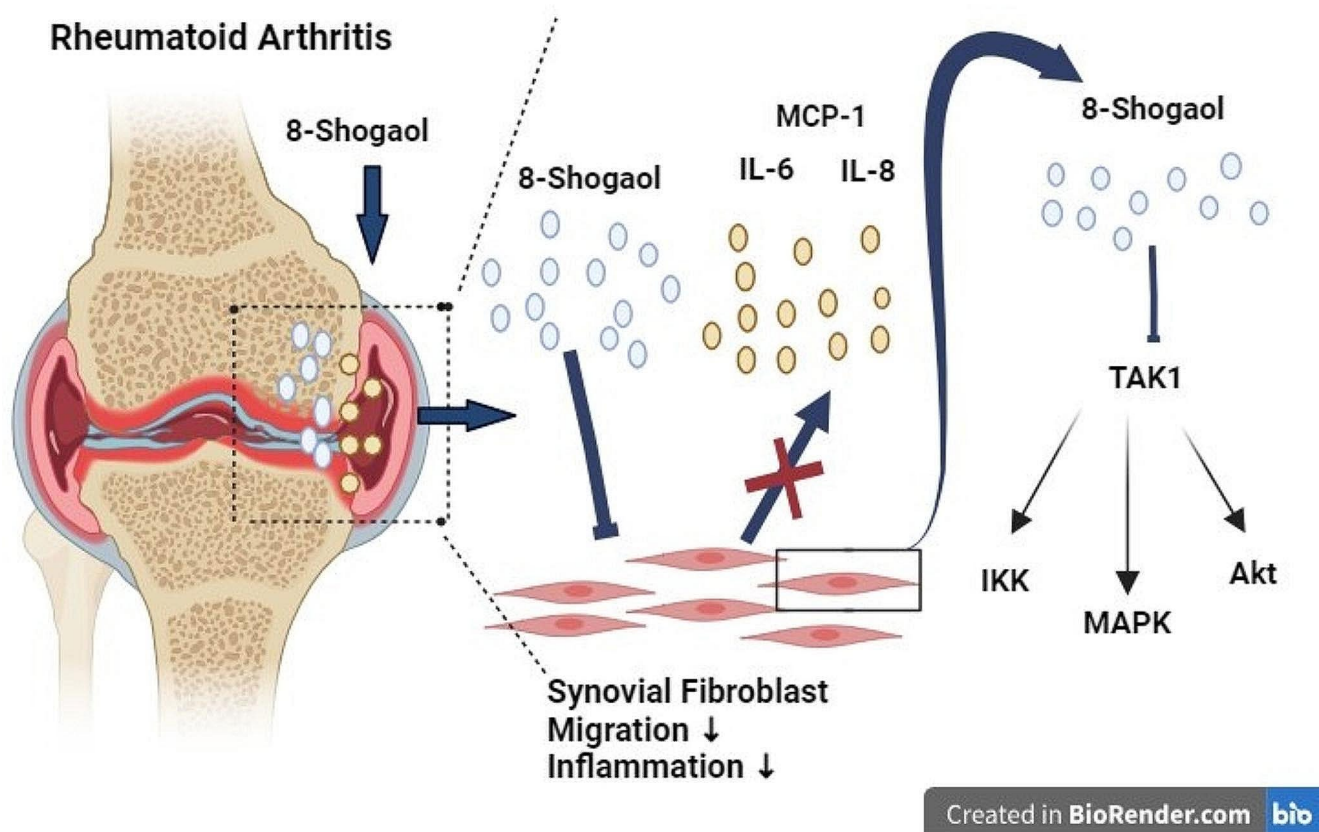


Fig. 6 Role of 8- Shogaol against Rheumatoid Arthritis by targeting TAK1 activity

stress generated during the onset of arthritic symptoms [46, 86].

Tripterygium wilfordii

Traditional Chinese herb *Tripterygium wilfordii* Hook F. (TwHF) is commonly grown in Korea, south and east of China, and Japan [87]. It has been used for a very long time in traditional Chinese medicine (TCM), frequently for rheumatoid arthritis [88]. TwHF has demonstrated a variety of pharmacological effects, including anticancer, anti-inflammatory, immuno-suppressive, anti-oxidative, and antifertility effects [89, 90]. Many preclinical studies have specifically shown cartilage-protective, immuno-suppressive, and anti-inflammatory benefits concerning its treatment for RA [91]. TwHF contains more than 500 isolated and recognized chemical components, including glycosides, sesquiterpenes, lignans, diterpenes, alkaloids, triterpenes, and flavonoids [92, 93]. Activating collaterals and blood circulation, reducing pain and swelling, and preventing rheumatism are just a few of the advantages of using TwHF, an herb that is said to be effective in arthralgia relief, according to traditional Chinese medicine (TCM) theory. A growing body of clinical research has demonstrated that TwHF-based therapy may be more successful than DMARD monotherapy at reducing the severity of RA illness [94, 95]. Following TwHF-based therapy, several patients may experience a variety of negative consequences, although these effects can be mitigated by using the proper treatment and management techniques [96]. Furthermore, Lv *et al.* recently discovered that MTX with TwHF therapy for treating RA was superior to both TwHF and MTX monotherapies [97]. As a result, more extensive clinical trials are still required to look into the safety, long-term effectiveness, and ideal dosages of using TwHF and MTX together to treat RA [98].

Boswellia serrata

The resin oleo-gum of the *Boswellia serrata* tree is one such folk remedy that has long been utilized for cosmetic, medicinal, and ceremonial purposes. The plant's oleo-gum resin is also used in traditional medicine to treat a wide range of ailments, such as inflammatory diseases like chronic pain, asthma, gastrointestinal issues, arthritis, and many other illnesses [99, 100]. Additionally, *Boswellia serrata* is mentioned throughout the Astanga Samgraha, Charaka Samhita, Astanga Hridaya, and Susruta Samhita original texts [101]. The oleo-gum resins of the tree have been described as an antiarthritic agent in the Susruta Samhita and Charaka Samhita. Additionally, it has been suggested for usage both externally and internally in several inflammatory disorders by the Unani and Ayurvedic medical systems [100].

Boswellia serrata acts to inhibit pro-inflammatory cytokines. It suppresses interferon- γ (IFN- γ), interleukin-1 β (IL-1 β), tumor necrosis factor- α (TNF- α), and increases the production of Interleukin-10 in collagen-induced arthritic rats. These cytokines have a crucial role in tissue damage during the progression of RA and chronic inflammation [81, 102]. Flavonoids, terpenoids, phenylpropanoids, and phenolic compounds were found to be the primary components in the gum resin of *B. serrata*, according to a phytochemical analysis using thin-layer chromatography [103]. *B. serrata* extract contains more than 12 distinct boswellic acids, but only AKBA (3-O-acetyl-11-keto-boswellic acid) and KBA (11-keto-boswellic acid) have attracted significant pharmacological interest [104]. *B. serrata* also contains additional elements that are crucial to its anti-inflammatory properties, including flavonoids (kaempferol, quercetin) and phenolic chemicals [105] Table 1.

Conclusion

In conclusion, this review has described a comprehensive overview that using herbal plants to cure arthritis is more effective than synthetic medications. The effects of plant extracts containing phyto-constituents such as terpenoids, flavonoids, sterols, and alkaloids that have anti-inflammatory and antioxidant action also have been assessed. As a result, herbal medicine is increasingly preferred as a complementary and alternative therapy because it has fewer or no negative effects. In vivo, in vitro, and clinical studies on RA have revealed information about the analgesic, anti-inflammatory, immune-modulatory, and antioxidant properties. To show the toxicity mechanisms and pharmacological interactions with herbal remedies, further focused investigations are required.

Limitations and future perspective

There are certain drawbacks to using these natural products, such as, low oral bioavailability, low efficacy, short half-life, and poor absorption because of rapid phytochemical metabolism. Excessive consumption of these natural products can also cause liver and renal toxicity, eye irritation, and skin irritation. To show the toxicity mechanisms and pharmacological interactions with herbal remedies, further focused investigations are required. The novel methods such as encapsulation, nano-micelles, nanoparticles, and lipid-core nano-capsules could be used to overcome these limitations. By using these methods, the efficacy, stability, and effective targeted delivery could be improved. So, these creative approaches may open the door to the development of brand-new all-natural RA drugs.

Table 1 Medicinal plants and their phytochemicals for the treatment of Arthritis

Medicinal Plant	Model	Effects	References
<i>Zingiber officinale Roscoe</i>	SCW method was used to induce arthritis in Female Lewis rats.	• Suppressing cytokine activity • Significantly reduction in joint swelling	[106]
<i>Zingiber officinale Roscoe</i>	Arthritic Lewis rats	• Significantly reduces the levels of important pro-inflammatory cytokines such as Interleukin-23, Interleukin-1, Interleukin-6, Interleukin-17, and chemokines MCP-1, TNF-α and MIP-1 (p 0.05).	[79]
<i>Curcuma longa</i>	Rodents	• Anti-inflammatory activity.	[107]
<i>Curcuma longa</i> ethanolic extract	Albino rats	• Arthritis prevention due to its anti-inflammatory properties.	[108]
<i>Curcuma longa</i> aqueous extract	Male albino mice	• Showed anti-inflammatory activity	[109]
Turmeric extract	Collagen-induced arthritic rats	• Reduced the deteriorating effects on the joints and bones.	[110]
<i>Nigella sativa</i> oil	FCA-induced arthritic rats	• Paw volume was significantly reduced when compared to the control group CFA. • Considerable anti-nociceptive activity in the oblique hind paw in comparison to the control group CFA.	[111]
Thymoquinone (TQ)	Human RA synovial fibroblasts	• Stabilizing Mcl-1 expression to induce apoptosis. • Inhibition of TNF-α-induced Interleukin-6 and Interleukin-8 production in a dose-dependent manner. • Decrease in TNF-α-induced ICAM-1, Cad-11 expression, and VCAM-1. • Inhibition of TNF-α-induced phosphorylation of JNK and p38 in a dose-dependent pattern. • TNF-α signaling is slowed down by preventing ASK1's phosphorylation and subsequent activation.	[112]
<i>Withania somnifera</i>	Rats	• Inflammation, pannus formation, and vascularity are all decreased by <i>withania somnifera</i>'s anti-arthritic effects.	[113]
<i>Withania somnifera</i>	Rats	• Attributing an anti-arthritic effect and suppressing the rheumatoid symptoms. • The inhibition of NF-kB activation and its controlled gene expression has a significant anti-inflammatory effect.	[63]
<i>Acori graminei</i> rhizoma	CIA model	• Lowering of serum type II collagen IgG levels and reduction of increased blood inflammation markers including Interleukin-6 and Tumor necrosis factor-α. • Both the edema of the hind limb and inflammatory cell infiltration into the synovial membrane decreased.	[114]
<i>Ribes orientale</i> extract	FCA-injected arthritic rats	• Reduced paw edema • Prevented body weight loss • Increase IL-10 and IL-4 levels while decreasing IL-1, TNF-α, IL-6, and NF-kB levels	[115]
Ethanolic leave extract of <i>S. chirayita</i>	Arthritic rats	• Reduces paw edema and pro-inflammatory cytokines including TNF-α and IL-1 in rats with arthritis.	[116]

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